

Introduction to Causal Inference and its applications

Hoang Van Ha
hvha@hcmus.edu.vn
University of Science, VNU-HCM

September, 2021

Outline

- 1 Introduction
- 2 Why causal inference?
- 3 Causal graphs and Structural Equation Models
 - Causal graphs
 - Structural Equation Models (SEMs)
- 4 Related Topics and Applications
 - Building and Evaluation of Impact of a recommender system
 - Unbiased Scene Graph Generation
 - Data-fusion Problem

Outline

- 1 Introduction
- 2 Why causal inference?
- 3 Causal graphs and Structural Equation Models
 - Causal graphs
 - Structural Equation Models (SEMs)
- 4 Related Topics and Applications
 - Building and Evaluation of Impact of a recommender system
 - Unbiased Scene Graph Generation
 - Data-fusion Problem

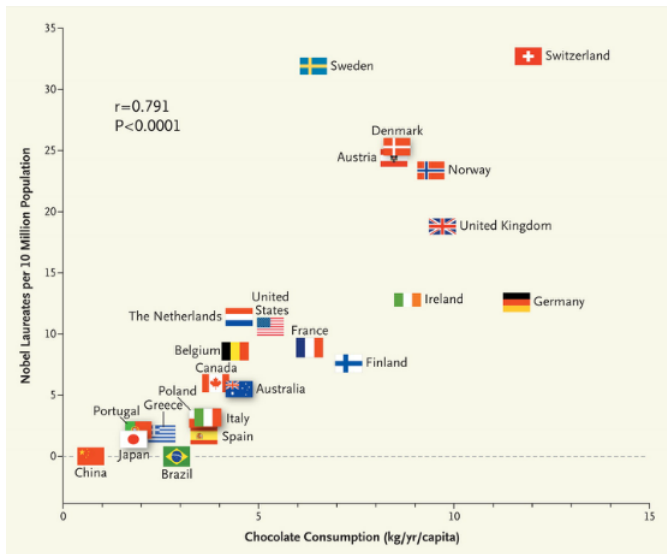
Association vs Causation

- **Association/prediction:** one of the main goals in most scientific/data analyses is to build a predictive model for establishing the correlation between an event and an outcome, *i.e* determine predictors or variables associated with the outcome of interest.

Association vs Causation

- **Association/prediction:** one of the main goals in most scientific/data analyses is to build a predictive model for establishing the correlation between an event and an outcome, *i.e* determine predictors or variables associated with the outcome of interest.
- But **association** does not imply **causation**.

Chocolate consumption vs Nobel prizes



Cancer and Red meats

LIVING



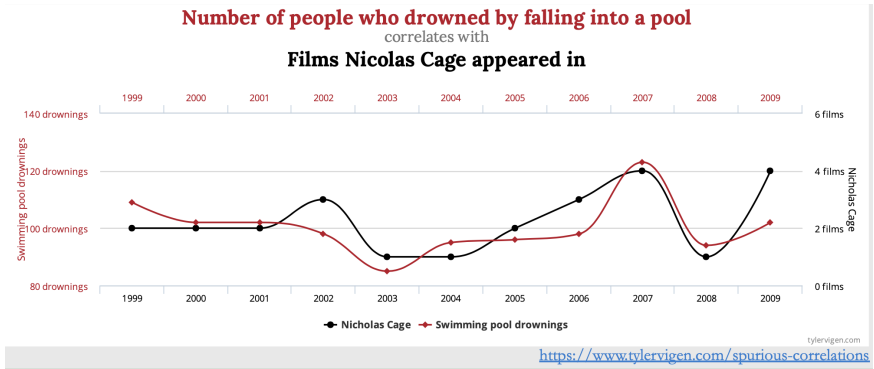
Scientists just pinpointed link between red meat and cancer

By Molly Shea

December 19, 2017 | 3:46pm | Updated



Nicolas Cage drives people to drown themselves



What is causal inference

- **Causality:** understand factors that cause or have an effect on the outcome of interest.

What is causal inference

- **Causality:** understand factors that cause or have an effect on the outcome of interest.
- Example:

What is causal inference

- **Causality:** understand factors that cause or have an effect on the outcome of interest.
- Example:
 - Effect of treatment on a disease

What is causal inference

- **Causality:** understand factors that cause or have an effect on the outcome of interest.
- Example:
 - Effect of treatment on a disease
 - Effect of climate change policy on emissions

What is causal inference

- **Causality:** understand factors that cause or have an effect on the outcome of interest.
- Example:
 - Effect of treatment on a disease
 - Effect of climate change policy on emissions
 - Effect of education on income

What is causal inference

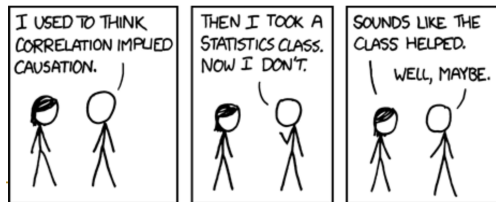
- **Causality:** understand factors that cause or have an effect on the outcome of interest.
- Example:
 - Effect of treatment on a disease
 - Effect of climate change policy on emissions
 - Effect of education on income
 - In general, effect of X on Y

What is causal inference

- **Causality:** understand factors that cause or have an effect on the outcome of interest.
- Example:
 - Effect of treatment on a disease
 - Effect of climate change policy on emissions
 - Effect of education on income
 - In general, effect of X on Y
- Recall: **association is not causation.**
When we observe X , we can predict that Y will occur $\neq Y$ occurs because of X .

What is causal inference

- **Causality:** understand factors that cause or have an effect on the outcome of interest.
- Example:
 - Effect of treatment on a disease
 - Effect of climate change policy on emissions
 - Effect of education on income
 - In general, effect of X on Y
- Recall: **association is not causation.**
When we observe X , we can predict that Y will occur $\neq Y$ occurs because of X .



Three Types of Analytics and Their Link to Causality ¹

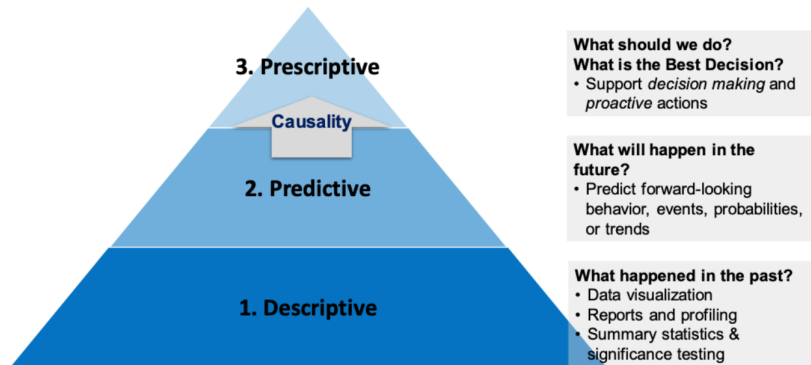


Figure 1. Three Types of Analytics

¹Top 10 Essential Data Science Topics to Real-World Application from the Industry Perspectives, see [here](#)

Causal Inference vs Machine Learning ²

- (Partly) supervised learning:
 - Assess causal effects: is this drug effective?
 - Assess mediation: is smoking bad for you because of smoke or because of nicotine?
 - Missing data: estimate the mean of the whole population from a biased sample.
- Unsupervised learning:
 - Learn "does A cause B?" from data.
 - Connections to learning graphical models from data.
- Decision making:
 - Agent learns to maximize causal effect of actions from data.
 - Connections to reinforcement learning.
- Models defined on **factual variables** (what we see), and **counterfactual variables** (what we don't see, but are interested in).

²This slide is taken from Causal Inference course of Ilya Shpitser (Johns Hopkins Univ.), available [here](#)

Outline

- 1 Introduction
- 2 Why causal inference?
- 3 Causal graphs and Structural Equation Models
 - Causal graphs
 - Structural Equation Models (SEMs)
- 4 Related Topics and Applications
 - Building and Evaluation of Impact of a recommender system
 - Unbiased Scene Graph Generation
 - Data-fusion Problem

Simpson's paradox

Example 2.1

We record the recovery rates of 700 patients who were given access to the drug. A total of 350 patients chose to take the drug and 350 patients did not. The results of the study are shown in Table

	Drug	No drug
Men	81 out of 87 recovered (93%)	234 out of 270 recovered (87%)
Women	192 out of 263 recovered (73%)	55 out of 80 recovered (69%)
Combined data	273 out of 350 recovered (78%)	289 out of 350 recovered (83%)

Simpson's paradox

Example 2.1

We record the recovery rates of 700 patients who were given access to the drug. A total of 350 patients chose to take the drug and 350 patients did not. The results of the study are shown in Table

	Drug	No drug
Men	81 out of 87 recovered (93%)	234 out of 270 recovered (87%)
Women	192 out of 263 recovered (73%)	55 out of 80 recovered (69%)
Combined data	273 out of 350 recovered (78%)	289 out of 350 recovered (83%)

Question: is the drug helpful?

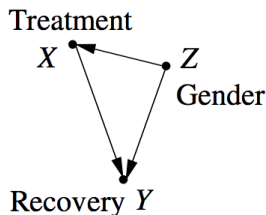
Simpson's paradox

	Drug	No drug
Men	81 out of 87 recovered (93%)	234 out of 270 recovered (87%)
Women	192 out of 263 recovered (73%)	55 out of 80 recovered (69%)
Combined data	273 out of 350 recovered (78%)	289 out of 350 recovered (83%)

Simpson's paradox

	Drug	No drug
Men	81 out of 87 recovered (93%)	234 out of 270 recovered (87%)
Women	192 out of 263 recovered (73%)	55 out of 80 recovered (69%)
Combined data	273 out of 350 recovered (78%)	289 out of 350 recovered (83%)

Causal graph:



Confounding: gender.

Main sources of influence ³

- ④ **Computer science** branch has been led by Judea Pearl: Directed Acyclic Graphs (DAGs) approach.

³This slide is taken from [here](#).

Main sources of influence ³

- ① **Computer science** branch has been led by Judea Pearl: Directed Acyclic Graphs (DAGs) approach.
- ② **Statistics:** most popular framework is known as **Potential Outcomes** framework and was proposed by Jerzy Neyman. This was the starting point for developing causal inference from a more statistical point of view.

³ This slide is taken from [here](#).

Main sources of influence ³

- ① **Computer science** branch has been led by Judea Pearl: Directed Acyclic Graphs (DAGs) approach.
- ② **Statistics:** most popular framework is known as **Potential Outcomes** framework and was proposed by Jerzy Neyman. This was the starting point for developing causal inference from a more statistical point of view.
- ③ **Econometrics:** techniques more popular among econometricians are:

³ This slide is taken from [here](#).

Main sources of influence ³

- ❶ **Computer science** branch has been led by Judea Pearl: Directed Acyclic Graphs (DAGs) approach.
- ❷ **Statistics:** most popular framework is known as **Potential Outcomes** framework and was proposed by Jerzy Neyman. This was the starting point for developing causal inference from a more statistical point of view.
- ❸ **Econometrics:** techniques more popular among econometricians are:
 - Controlled Regression

³ This slide is taken from [here](#).

Main sources of influence ³

- ❶ **Computer science** branch has been led by Judea Pearl: Directed Acyclic Graphs (DAGs) approach.
- ❷ **Statistics:** most popular framework is known as **Potential Outcomes** framework and was proposed by Jerzy Neyman. This was the starting point for developing causal inference from a more statistical point of view.
- ❸ **Econometrics:** techniques more popular among econometricians are:
 - Controlled Regression
 - Regression Discontinuity Design

³ This slide is taken from [here](#).

Main sources of influence ³

- ❶ **Computer science** branch has been led by Judea Pearl: Directed Acyclic Graphs (DAGs) approach.
- ❷ **Statistics:** most popular framework is known as **Potential Outcomes** framework and was proposed by Jerzy Neyman. This was the starting point for developing causal inference from a more statistical point of view.
- ❸ **Econometrics:** techniques more popular among econometricians are:
 - Controlled Regression
 - Regression Discontinuity Design
 - Difference-in-Differences

³ This slide is taken from [here](#).

Main sources of influence ³

- ❶ **Computer science** branch has been led by Judea Pearl: Directed Acyclic Graphs (DAGs) approach.
- ❷ **Statistics:** most popular framework is known as **Potential Outcomes** framework and was proposed by Jerzy Neyman. This was the starting point for developing causal inference from a more statistical point of view.
- ❸ **Econometrics:** techniques more popular among econometricians are:
 - Controlled Regression
 - Regression Discontinuity Design
 - Difference-in-Differences
 - Fixed-Effects Regression

³ This slide is taken from [here](#).

Main sources of influence ³

- ❶ **Computer science** branch has been led by Judea Pearl: Directed Acyclic Graphs (DAGs) approach.
- ❷ **Statistics:** most popular framework is known as **Potential Outcomes** framework and was proposed by Jerzy Neyman. This was the starting point for developing causal inference from a more statistical point of view.
- ❸ **Econometrics:** techniques more popular among econometricians are:
 - Controlled Regression
 - Regression Discontinuity Design
 - Difference-in-Differences
 - Fixed-Effects Regression
 - Instrumental Variables

³ This slide is taken from [here](#).

Pearl's ladder of causation

- **Level 1. Association:** One object is associated with another if observing one changes the probability of observing the other

Pearl's ladder of causation

- **Level 1. Association:** One object is associated with another if observing one changes the probability of observing the other $\rightarrow$ *Observational studies*.

Pearl's ladder of causation

- **Level 1. Association:** One object is associated with another if observing one changes the probability of observing the other $\rightarrow$ *Observational studies*.
- **Level 2. Intervention:** This level asserts specific causal relationships between events

Pearl's ladder of causation

- **Level 1. Association:** One object is associated with another if observing one changes the probability of observing the other $\rightarrow$ *Observational studies*.
- **Level 2. Intervention:** This level asserts specific causal relationships between events $\rightarrow$ *Randomized control trials*.

Pearl's ladder of causation

- **Level 1. Association:** One object is associated with another if observing one changes the probability of observing the other → *Observational studies*.
- **Level 2. Intervention:** This level asserts specific causal relationships between events → *Randomized control trials*.
- **Level 3. Counterfactual:** involves constructing a theory of the world that explains why specific actions have specific effects and what happens in the absence of such actions.

Pearl's ladder of causation

Level (Symbol)	Typical Activity	Typical Questions	Examples
1. Association $P(y x)$	Seeing	What is? How would seeing X change my belief in Y ?	What does a symptom tell me about a disease? What does a survey tell us about the election results?
2. Intervention $P(y do(x), z)$	Doing Intervening	What if? What if I do X ?	What if I take aspirin, will my headache be cured? What if we ban cigarettes?
3. Counterfactuals $P(y_x x', y')$	Imagining, Retrospection	Why? Was it X that caused Y ? What if I had acted differently?	Was it the aspirin that stopped my headache? Would Kennedy be alive had Oswald not shot him? What if I had not been smoking the past 2 years?

Figure: The Causal Hierarchy (Table from Pearl, 2018).

Intervention: Do-Calculus

Name: The Causal Calculus (do-calculus, Pearl's Causal Calculus, Calculus of Actions)

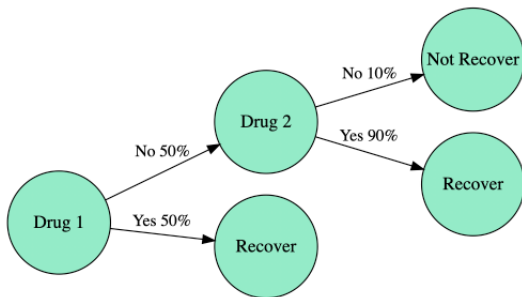
Shortly: Calculus to discuss causality in a formal language by Judea Pearl A new operator, $\text{do}()$, marks an action or an intervention in the model. In an algebraic model we replace certain functions with a constant $X = x$.

The causal calculus uses Bayesian conditioning, $p(y|X = x)$, where x is observed variable, and causal conditioning, $p(y|\text{do}(X = x))$, where an action is taken to force a specific value x .

Goal is to generate probabilistic formulas for the effect of interventions in terms of the observed probabilities.

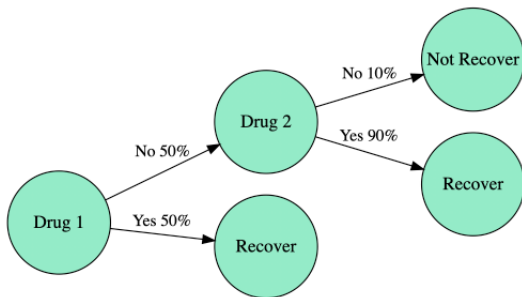
Observing vs intervening: example

This example is taken from [here](#)



Observing vs intervening: example

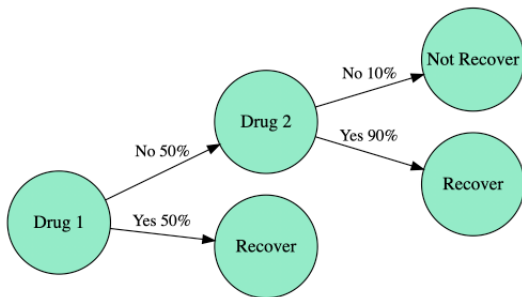
This example is taken from [here](#)



Question: What if the laboratory stops manufacturing Drug 2? What is our probability of recovery?

Observing vs intervening: example

This example is taken from [here](#)



Question: What if the laboratory stops manufacturing Drug 2? What is our probability of recovery?

Answer: Realistically 50%

Observing

Drug 1	Drug 2	Cured	Total
Yes	No	50	50
No	Yes	45	45
No	No	0	5

Observing

Drug 1	Drug 2	Cured	Total
Yes	No	50	50
No	Yes	45	45
No	No	0	5

If there is no Drug 2

Drug 1	Drug 2	Cured	Total
Yes	No	50	50
No	No	0	5

Observing

Drug 1	Drug 2	Cured	Total
Yes	No	50	50
No	Yes	45	45
No	No	0	5

If there is no Drug 2

Drug 1	Drug 2	Cured	Total
Yes	No	50	50
No	No	0	5

$$P(\text{Recovering} \mid \text{Drug 2} = \text{No}) = 50 / (50 + 5) \approx 91\%$$

Observing vs intervening

What if the laboratory stops manufacturing Drug 2?

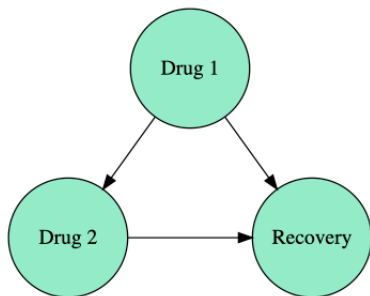


Figure: Full graph

Observing vs intervening

What if the laboratory stops manufacturing Drug 2?

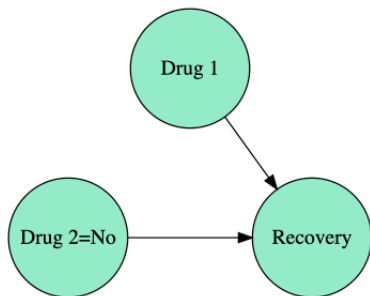
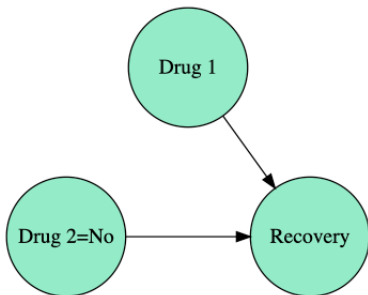


Figure: Interventional graph

Observing is not intervening



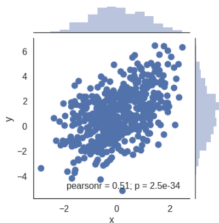
Drug 1	Drug 2	Cured	Total
Yes	No	50	50
No	No	0	50

$$P(\text{Recovering} | \text{do}(\text{Drug2} = \text{No})) = 50 / (50 + 50) = 50\%$$

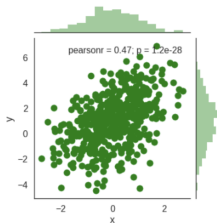
Observing vs. interventioning: a toy example ⁴

Suppose that we have three scripts that would implement the same, correct sampling behaviour:

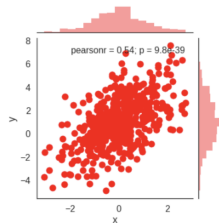
```
x = randn()
y = x + 1 + sqrt(3)*randn()
```



```
y = 1 + 2*randn()
x = (y-1)/4 + sqrt(3)*randn()/2
```



```
z = randn()
y = z + 1 + sqrt(3)*randn()
x = z
```

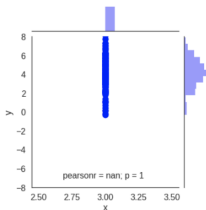


⁴ Example of Ferenc Huszár, available [here](#)

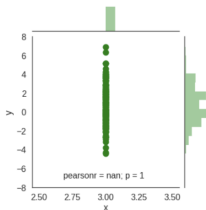
Observing vs. interventioning: a toy example

Intervention: suppose that a hacker insert the line $x = 3$ ($\text{do}(x = 3)$) after each line of the code:

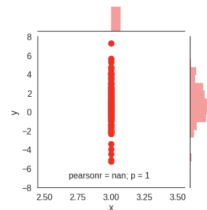
```
x = randn()
x = 3
y = x + 1 + sqrt(3)*randn()
x = 3
```



```
y = 1 + 2*randn()
x = 3
x = (y-1)/4 + sqrt(3)*randn()/2
x = 3
```

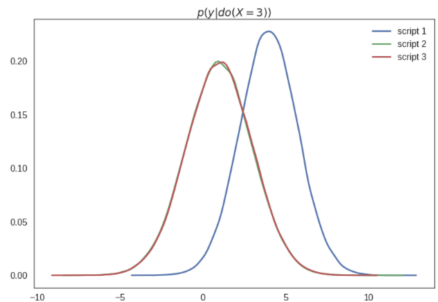
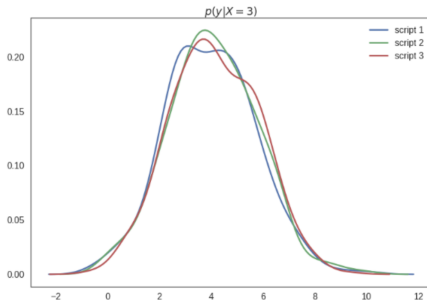


```
z = randn()
x = 3
x = z
x = 3
y = z + 1 + sqrt(3)*randn()
x = 3
```



Observing vs. interventioning: a toy example

The scripts behave differently under intervention:



Outline

- 1 Introduction
- 2 Why causal inference?
- 3 Causal graphs and Structural Equation Models
 - Causal graphs
 - Structural Equation Models (SEMs)
- 4 Related Topics and Applications
 - Building and Evaluation of Impact of a recommender system
 - Unbiased Scene Graph Generation
 - Data-fusion Problem

Outline

- 1 Introduction
- 2 Why causal inference?
- 3 Causal graphs and Structural Equation Models
 - Causal graphs
 - Structural Equation Models (SEMs)
- 4 Related Topics and Applications
 - Building and Evaluation of Impact of a recommender system
 - Unbiased Scene Graph Generation
 - Data-fusion Problem

Basic Elements of Causal Graphs/DAGs

- **DAGs:** Directed Acyclic Graphs (there is no cycle in the graphs)

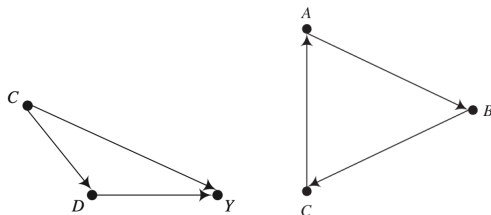


Figure: A DAG (left) and a directed graph that includes a cycle.

Basic Elements of Causal Graphs/DAGs

- **DAGs:** Directed Acyclic Graphs (there is no cycle in the graphs)
- Start with a set of events/random variables V

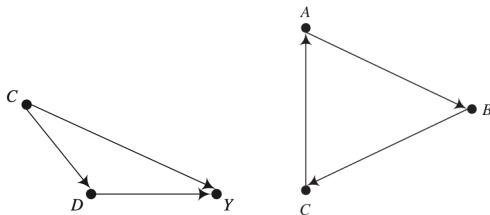


Figure: A DAG (left) and a directed graph that includes a cycle.

Basic Elements of Causal Graphs/DAGs

- **DAGs:** Directed Acyclic Graphs (there is no cycle in the graphs)
- Start with a set of events/random variables V
- Each element of V is a vertex in causal graph G

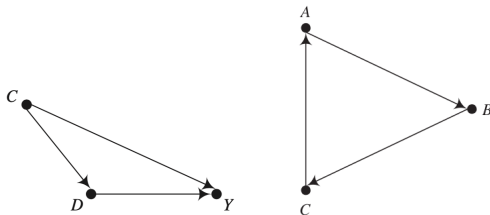


Figure: A DAG (left) and a directed graph that includes a cycle.

Basic Elements of Causal Graphs/DAGs

- **DAGs:** Directed Acyclic Graphs (there is no cycle in the graphs)
- Start with a set of events/random variables V
- Each element of V is a vertex in causal graph G
- Causal effect: single-headed arrow $\longrightarrow$. A causes B is causal graph G only if A is an ancestor of B

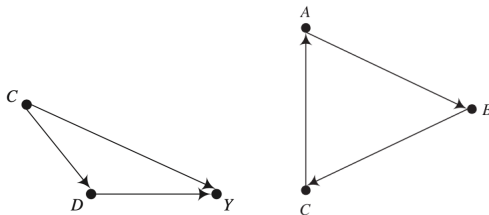


Figure: A DAG (left) and a directed graph that includes a cycle.

Basic Elements of Causal Graphs/DAGs

- **DAGs:** Directed Acyclic Graphs (there is no cycle in the graphs)
- Start with a set of events/random variables V
- Each element of V is a vertex in causal graph G
- Causal effect: single-headed arrow $\longrightarrow$. A causes B is causal graph G only if A is an ancestor of B
- DAGs with such an assumption are causal graphs

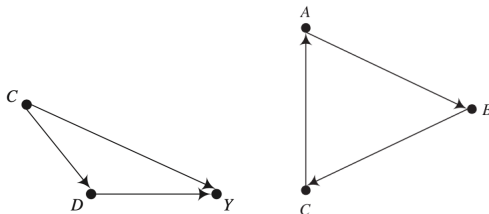
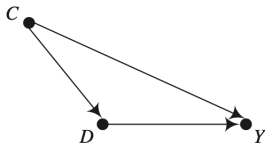
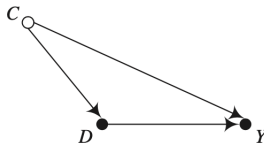


Figure: A DAG (left) and a directed graph that includes a cycle.

Basic Elements of Causal Graphs



(a)



(b)

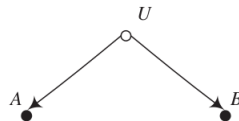
- Observed random variable •
- Unobserved random variable ○
- A *path* is a sequence of edges

- A *directed path* is a path with all edges pointing to the same direction
- A variable is a *descendant* of another variable if it can be reached by a directed path

Basic Elements of Causal Graphs



(a)



(b)

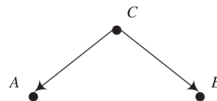
- Directed path of length one $A \longrightarrow B$, then A is the *parent* and B is the *child*.
- If two variables are mutually dependent on unobserved variables we can use curved, dashed bidirected arrow.

Causal Graphs for Three Variables

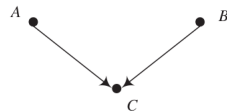
- (a) A and B are unconditionally associated
- (b) A and B are unconditionally associated but it's because they mutually depend on C
- (c) A and B are both causes of C, Pearl labels C as *collider variable*. There is a path from A to B but it does not generate unconditional association between A and B.



(a) Mediation

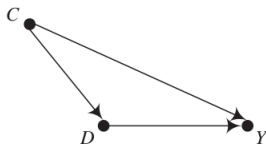


(b) Mutual dependence

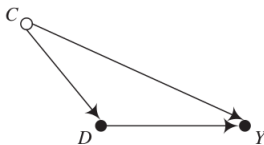


(c) Mutual causation

Confounding



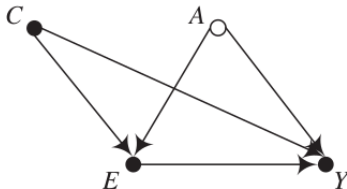
(a)



(b)

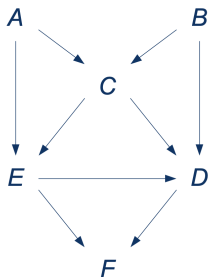
- Total association is composed by 2 pieces: (1) causal effect of D on Y ; (2) common dependence of D and Y on C . We say that the causal effect of D on Y is *confounded* by C , or C is a *confounder*.
- **Problem:** deconfounding!

Confounding: example



- Education, E , is a cause of earnings, Y , but this causal effect of E on Y is confounded by two sources: (1) observed variable C such as demographic characteristics and family background and (2) unobserved variable for ability, A .
- **Problem:** deconfounding!

Causal graphs: summary



- E is a child of A , F is a descendant of A
- F is a collider along $E \rightarrow F \leftarrow D$, C is a collider along $E \leftarrow A \rightarrow C \leftarrow B \rightarrow D$
- $A \rightarrow C \rightarrow D \rightarrow F$ is a directed path from A to F
- $E \leftarrow A \rightarrow C \rightarrow D \rightarrow F$ is a path from E to F

DAGs and Probability distributions

DAGs encode assumptions about dependencies between variables (nodes). A DAG will tell us:

- which variables are **independent** from each other
- which variables are **conditionally independent** from each other
- ways that we can **factor and simplify** the joint distribution

DAGs and Probability distributions

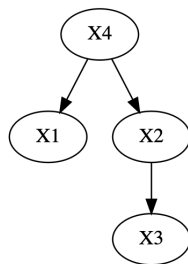
DAGs encode assumptions about dependencies between variables (nodes). A DAG will tell us:

- which variables are **independent** from each other
- which variables are **conditionally independent** from each other
- ways that we can **factor and simplify** the joint distribution

Local Markov Assumption

Let G be a DAG representing a causal system over a set of r.v's V , and P a distribution over V . Given its parents in G , a node/variable X is independent of all its non-descendants.

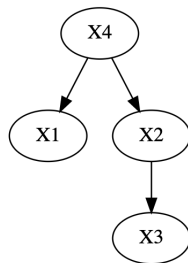
DAGs and Probability Distributions



This DAG implies:

- $X_1 \perp\!\!\!\perp X_2, X_3 | X_4 \Rightarrow P(x_1 | x_2, x_3, x_4)$

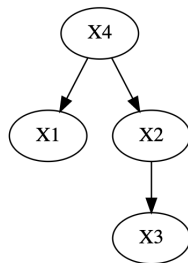
DAGs and Probability Distributions



This DAG implies:

- $X_1 \perp\!\!\!\perp X_2, X_3 | X_4 \Rightarrow P(x_1 | x_2, x_3, x_4)$

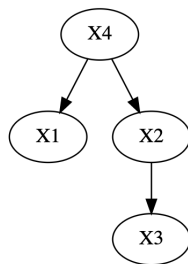
DAGs and Probability Distributions



This DAG implies:

- $X_1 \perp\!\!\!\perp X_2, X_3 | X_4 \Rightarrow P(x_1 | x_2, x_3, x_4) = P(x_1 | x_4)$

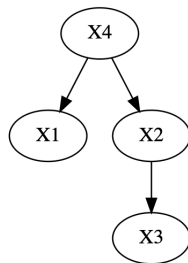
DAGs and Probability Distributions



This DAG implies:

- $X_1 \perp\!\!\!\perp X_2, X_3 | X_4 \Rightarrow P(x_1 | x_2, x_3, x_4) = P(x_1 | x_4)$
- $X_4 \perp\!\!\!\perp X_3 | X_2 \Rightarrow P(x_4 | x_1, x_2, x_3)$

DAGs and Probability Distributions



This DAG implies:

- $X_1 \perp\!\!\!\perp X_2, X_3 | X_4 \Rightarrow P(x_1 | x_2, x_3, x_4) = P(x_1 | x_4)$
- $X_4 \perp\!\!\!\perp X_3 | X_2 \Rightarrow P(x_4 | x_1, x_2, x_3) = P(x_4 | x_1, x_2)$

Decomposition of Joint Distribution

Chain rule:

$$P(x_1, \dots, x_n) = \prod_j P(x_j | x_1, \dots, x_{j-1}).$$

With local Markov assumption

$$P(x_j | x_1, \dots, x_{j-1}) = P(x_j | \text{pa}_j),$$

where pa_j is a minimal set of predecessors of X_j that renders X_j independent of all its other predecessors.

Thus

$$P(x_1, \dots, x_n) = \prod_j P(x_j | \text{pa}_j).$$

Decomposition of Joint Distribution

Chain rule:

$$P(x_1, \dots, x_n) = \prod_j P(x_j | x_1, \dots, x_{j-1}).$$

With local Markov assumption

$$P(x_j | x_1, \dots, x_{j-1}) = P(x_j | \text{pa}_j),$$

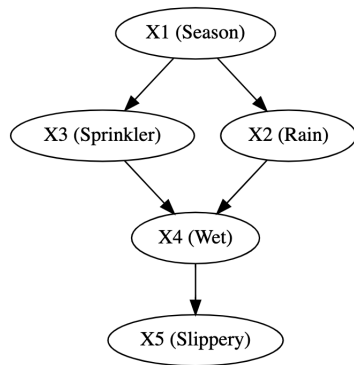
where pa_j is a minimal set of predecessors of X_j that renders X_j independent of all its other predecessors.

Thus

$$P(x_1, \dots, x_n) = \prod_j P(x_j | \text{pa}_j).$$

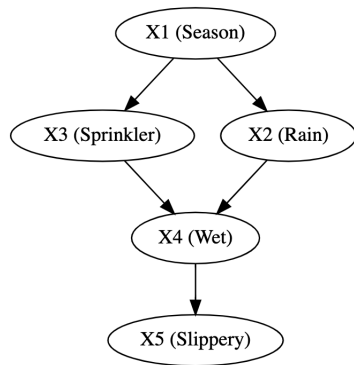
⇒ Bayesian network factorization

Decomposition Example



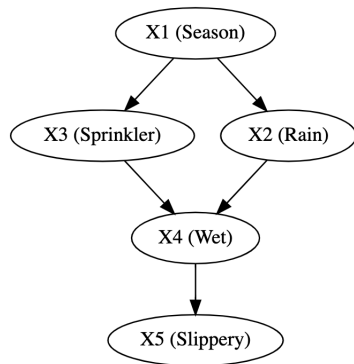
$$P(x_1, x_2, x_3, x_4, x_5) = P(x_1)P(x_2|x_1)P(x_3|x_1, x_2)P(x_4|x_1, x_2, x_3) \\ \times P(x_5|x_1, x_2, x_3, x_4)$$

Decomposition Example



$$P(x_1, x_2, x_3, x_4, x_5) = P(x_1)P(x_2|x_1)P(x_3|x_1, x_2)P(x_4|x_1, x_2, x_3) \\ \times P(x_5|x_1, x_2, x_3, x_4)$$

Decomposition Example



$$\begin{aligned} P(x_1, x_2, x_3, x_4, x_5) &= P(x_1)P(x_2|x_1)P(x_3|x_1, x_2)P(x_4|x_1, x_2, x_3) \\ &\quad \times P(x_5|x_1, x_2, x_3, x_4) \\ &= P(x_1)P(x_2|x_1)P(x_3|x_1)P(x_4|x_2, x_3)P(x_5|x_4). \end{aligned}$$

Compatibility between DAGs and Distributions

Definition 3.1

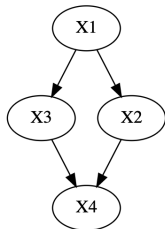
If a probability function P admits the factorization

$$P(x_1, \dots, x_n) = \prod_j P(x_j \mid \text{pa}_j),$$

relative to DAG G , we say that G represents P , that G and P are compatible, or that P is Markov relative to G .

Compatibility between DAGs and Distributions

The DAG



admits this factorization

$$P(x_1, x_2, x_3, x_4) = P(x_1)P(x_2|x_1)P(x_3|x_1)P(x_4|x_2, x_3).$$

We say that this probability function and this DAG are compatible.

d -separation (conditional independence)

- **Blocking:** paths can be blocked by conditioning on nodes in the path.
- Consider the path $A \rightarrow B \rightarrow C$. If we condition on B (a node in the middle of the path), we block the path from A to C .
- **Blocking with colliders:** consider the path $A \rightarrow B \leftarrow C$. Here, B is a collider and A and C are not associated via this path. Information from A and C collides at B , thus the path is blocked by default. If we condition on B , this will induce an association between A and C .

Rules for d -separation

Definition 3.2

A path p is said to be d -separated (or blocked) by a set of nodes Z if and only if

- 1) p contains a chain $i \rightarrow m \rightarrow j$ or a fork $i \leftarrow m \rightarrow j$ such that the middle node m is in Z , or
- 2) p contains an inverted fork (or collider) $i \rightarrow m \leftarrow j$ such that the middle node m is not in Z and such that no descendant of m is in Z .

A set Z is said to d -separated X from Y if and only if Z blocks every path from a node in X to a node in Y .

d -separation example

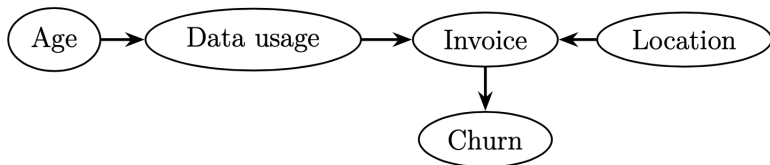


Figure: A causal graph for churn analysis on telecom customer data, illustrating the notion d -separation

- Age and Invoice are d -separated given Data usage.
- Data usage and Location are no longer d -separated once we know the invoice amount or the churn variable.

Outline

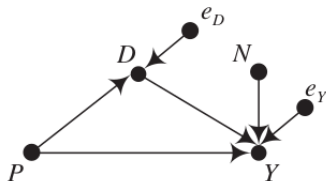
- 1 Introduction
- 2 Why causal inference?
- 3 Causal graphs and Structural Equation Models
 - Causal graphs
 - Structural Equation Models (SEMs)
- 4 Related Topics and Applications
 - Building and Evaluation of Impact of a recommender system
 - Unbiased Scene Graph Generation
 - Data-fusion Problem

Structural Equation Models (SEMs)

- ① Traditional linear additive path models
- ② Nonparametric structural Equations

Example: Charter school

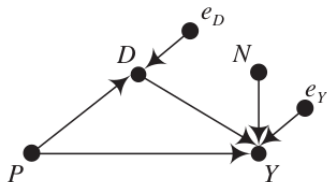
Question: Do students who attend charter schools perform better on standardized tests than they would have performed if they had instead attended regular public schools?



A study about the attending charter schools and test scores.

- Y is the test score
- D is whether or not the student attended a charter school last year
- P represents parental background
- N represents neighborhood of residence
- e_D and e_Y are other variables that have effects on D and Y

Linear additive path models

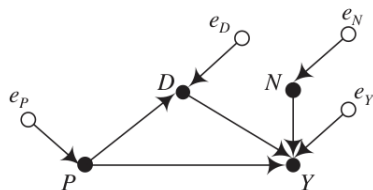


Linear additive model

$$D = a_D + b_P P + e_D$$

$$Y = a_Y + b_D D + b_P P + b_N N + e_Y$$

Nonparametric Structural Equations



$$P = f_P(e_P)$$

$$D = f_D(P, e_D)$$

$$N = f_N(e_N)$$

$$Y = f_Y(P, D, N, e_Y)$$

- the error terms represent independent background factors that the investigator choose not to include in the analysis
- these errors are always assumed to be independent of each other and of all other variables in the graph
- the error terms $e_{\bullet}$ can be suppressed in the standard representation

Outline

- 1 Introduction
- 2 Why causal inference?
- 3 Causal graphs and Structural Equation Models
 - Causal graphs
 - Structural Equation Models (SEMs)
- 4 Related Topics and Applications
 - Building and Evaluation of Impact of a recommender system
 - Unbiased Scene Graph Generation
 - Data-fusion Problem

Other topics that are not included in this talk

- Adjustment using Back-door and Front-door criteria
- Selection bias
- Instrumental variables
- Potential outcomes and counterfactual
- Causal discovery
- Causal Reinforcement Learning (= Causal Inference + Reinforcement Learning)

Outline

- 1 Introduction
- 2 Why causal inference?
- 3 Causal graphs and Structural Equation Models
 - Causal graphs
 - Structural Equation Models (SEMs)
- 4 Related Topics and Applications
 - Building and Evaluation of Impact of a recommender system
 - Unbiased Scene Graph Generation
 - Data-fusion Problem

Building a Recommendation Systems ⁵

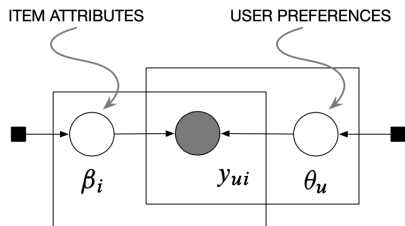
						
						
						
						
						

- In: Ratings or click data
- Out: A system that provides recommendations

⁵ These slide are taken from the presentation of David M. Blei, available [here](#).

Building a Recommendation Systems

Classical solution: Matrix factorization



$$\theta_u \sim f(\cdot)$$

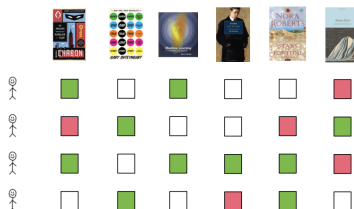
$$\beta_i \sim g(\cdot)$$

$$y_{ui} \sim \text{exp-fam}(\theta_u^\top \beta_i)$$

- ▶ Users have *latent preferences* θ_u .
Items have *latent attributes* β_i .
- ▶ A users rating comes from an exponential family,
e.g., Poisson, Bernoulli, Gaussian.

Building a Recommendation Systems

Approach with causal inference: Causal recommendation (David M. Blei, Dawen Liang, Laurent Charlin) ⁶.



- Treat the inference of **user preferences** as a causal inference
- We want preferences θ_i to answer (from attributes β_j):

What if user i saw item j ? How would user i like it?

- This leads to a different approach from classical matrix factorization.

⁶Y. Wang, D. Liang, L. Charlin, and D. Blei. Causal inference for recommender systems. ACM Conference on Recommender Systems, 2020.

Building a Recommendation Systems

- Suppose: Items = movies.

Building a Recommendation Systems

- Suppose: Items = movies.
- In prediction, the recommender system is trying to answer "How would the user rate this movie if she saw it?"

Building a Recommendation Systems

- Suppose: Items = movies.
- In prediction, the recommender system is trying to answer "How would the user rate this movie if she saw it?"
 - ▶ *Recommending all the movies that users will like or may not be the most cost-efficient strategy.*

Building a Recommendation Systems

- Suppose: Items = movies.
- In prediction, the recommender system is trying to answer "How would the user rate this movie if she saw it?"
 - ▶ *Recommending all the movies that users will like or may not be the most cost-efficient strategy.*
- Recommend only the movies that
 - (1) *if made exposed, the user will go see them*
 - (2) *if not, the user will not go see them.*

Building a Recommendation Systems

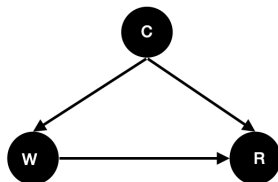
- Suppose: Items = movies.
- In prediction, the recommender system is trying to answer "How would the user rate this movie if she saw it?"
 - ▶ *Recommending all the movies that users will like or may not be the most cost-efficient strategy.*
- Recommend only the movies that
 - (1) *if made exposed, the user will go see them*
 - (2) *if not, the user will not go see them.*
 - ▶ **Interventional question:** *what would the rating be if we make the user exposed (or not exposed) to the movie?*

Building a Recommendation Systems

- For classical approach (Matrix Factorization): builds a model from observed ratings data then predict unseen ratings. This strategy only provides valid causal inferences if users randomly watched movies.
But users do not (usually) watch movies at random → answering the causal question from observed ratings data is challenging.

Building a Recommendation Systems

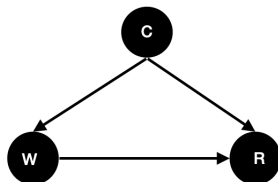
- For classical approach (Matrix Factorization): builds a model from observed ratings data then predict unseen ratings. This strategy only provides valid causal inferences if users randomly watched movies.
But users do not (usually) watch movies at random → answering the causal question from observed ratings data is challenging.
- Issue:** Confounders.



C = confounders that affect both the treatment assignments W (which movies the users watch) and the outcomes R (how they rate them).

Building a Recommendation Systems

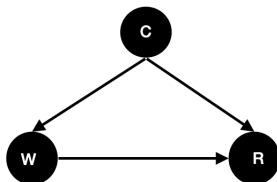
- For classical approach (Matrix Factorization): builds a model from observed ratings data then predict unseen ratings. This strategy only provides valid causal inferences if users randomly watched movies.
But users do not (usually) watch movies at random → answering the causal question from observed ratings data is challenging.
- Issue:** Confounders.



C = confounders that affect both the treatment assignments W (which movies the users watch) and the outcomes R (how they rate them).

Building a Recommendation Systems

- For classical approach (Matrix Factorization): builds a model from observed ratings data then predict unseen ratings. This strategy only provides valid causal inferences if users randomly watched movies.
But users do not (usually) watch movies at random → answering the causal question from observed ratings data is challenging.
- **Issue:** Confounders.



C = confounders that affect both the treatment assignments W (which movies the users watch) and the outcomes R (how they rate them).

⇒ **deconfounded recommender** (Y. Wang, D. Liang, L. Charlin, and D. Blei. Causal inference for recommender systems. ACM Conference on Recommender Systems, 2020.)

Application: Impact of a recommender system ⁷

- Estimating the causal impact the causal impact of Amazon.com's recommender system using Split-door criterion.

The Annals of Applied Statistics
2018, Vol. 12, No. 4, 2699–2733
<https://doi.org/10.1214/18-AOS1179>
© Institute of Mathematical Statistics, 2018

SPLIT-DOOR CRITERION: IDENTIFICATION OF CAUSAL EFFECTS THROUGH AUXILIARY OUTCOMES

BY AMIT SHARMA, JAKE M. HOFMAN AND DUNCAN J. WATTS

Microsoft Research

We present a method for estimating causal effects in time series data when fine-grained information about the outcome of interest is available. Specifically, we examine what we call the *split-door* setting, where the outcome variable can be split into two parts: one that is potentially affected by the cause being studied and another that is independent of it, with both parts sharing the same (unobserved) confounders. We show that under these conditions, the problem of identification reduces to that of testing for independence among observed variables, and propose a method that uses this approach to automatically find subsets of the data that are causally identified. We demonstrate the method by estimating the causal impact of Amazon's recommender system on traffic to product pages, finding thousands of examples within the dataset that satisfy the split-door criterion. Unlike past studies based on natural experiments that were limited to a single product category, our method applies to a large and representative sample of products viewed on the site. In line with previous work, we find that the widely-used click-through rate (CTR) metric overestimates the causal impact of recommender systems; depending on the product category, we estimate that 50–80% of the traffic attributed to recommender systems would have happened even without any recommendations. We conclude with guidelines for using the split-door criterion as well as a discussion of other contexts where the method can be applied.



Purity: A Novel Paperback – August 2, 2016
Customers Who Bought This Item Also Bought



FIG. 5. Screenshot of a focal product, the book “Purity,” and its recommendations on Amazon.com.

⁷ Sharma et al. Split-door criterion: identification of causal effects through auxiliary outcomes. *The Annals of Applied Statistics*, 2018.

Application: Impact of a recommender system

- Question: how exactly exposure to these recommended products changes consumer activity?
 - ▶ **Counterfactual question:** what would have happened had someone visited a focal product but had not been exposed to any recommendations?
 - ▶ How much traffic recommender systems *cause*, over and above what would have happened in their absence?

Application: Impact of a recommender system

- Question: how exactly exposure to these recommended products changes consumer activity?
 - ▶ **Counterfactual question:** what would have happened had someone visited a focal product but had not been exposed to any recommendations?
 - ▶ How much traffic recommender systems *cause*, over and above what would have happened in their absence?
- **Naive method:** assume that users would not have viewed these other products without the recommender system, and as a result simply compute the observed click-through rate on recommendations [Grau (2009), Mulpuru (2006)].

Application: Impact of a recommender system

- Question: how exactly exposure to these recommended products changes consumer activity?
 - ▶ **Counterfactual question:** what would have happened had someone visited a focal product but had not been exposed to any recommendations?
 - ▶ How much traffic recommender systems *cause*, over and above what would have happened in their absence?
- **Naive method:** assume that users would not have viewed these other products without the recommender system, and as a result simply compute the observed click-through rate on recommendations [Grau (2009), Mulpuru (2006)].
- **Solution:** use the split-door criterion to eliminate the issue of correlated demand by automatically identifying and analyzing instances where demand for a product and one (or more) of its recommendations are independent over some time period.
 - ▶ formalize this problem through a causal graphical model of recommender system traffic
 - ▶ reveal a structure amenable to the split-door criterion
 - ▶ apply the criterion to a large-scale dataset of web browsing activity on Amazon.com to discover thousands of instances satisfying the criterion

Application: Impact of a recommender system

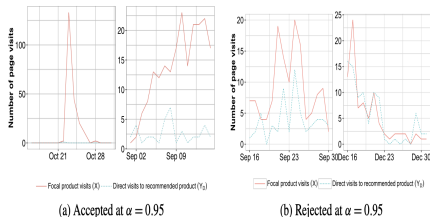


FIG. 6. Examples of time series for focal and recommended products that are (a) accepted or (b) rejected by the split-door criterion at a significance level of $\alpha = 0.95$ for the independence test.

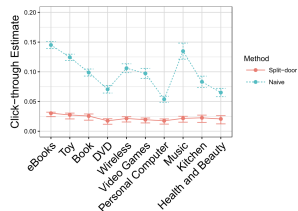


FIG. 8. Comparison of the causal CTR with the naive observational CTR for products that satisfy the split-door criterion. Categories are ordered by the number of products found by the split-door criterion in each category, with eBooks containing the most and Health and Beauty the least.

Results show that a naive observational estimate of the impact of this recommender system overstates the causal impact on the products analyzed by a factor of at least two.

Outline

- 1 Introduction
- 2 Why causal inference?
- 3 Causal graphs and Structural Equation Models
 - Causal graphs
 - Structural Equation Models (SEMs)
- 4 Related Topics and Applications
 - Building and Evaluation of Impact of a recommender system
 - **Unbiased Scene Graph Generation**
 - Data-fusion Problem

Application: Unbiased Scene Graph Generation (Tang et al. 2020)⁸

- Traditional machine learning debiasing methods cannot distinguish between good and bad bias. Example, good context prior ('person read book' rather than 'eat'), and bad long-tailed bias ('near' dominating 'behind / in front of').
- In this paper, the authors present a novel framework based on causal inference and build a causal graph to perform traditional biased training with the graph. This framework, the authors claim, is agnostic and thus, can be widely applied in the community who seeks unbiased predictions.

⁸Tang et al., Unbiased Scene Graph Generation from Biased Training. arXiv:2002.11949v3, 2020.

Application: Unbiased Scene Graph Generation (Tang et al. 2020)

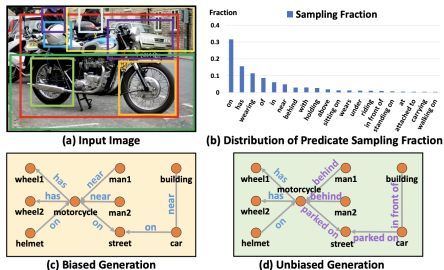


Figure 1. An example of scene graph generation (SGG). (a) An input image with bounding boxes. (b) The distribution of sample fraction for the most frequent 20 predicates in Visual Genome [22]. (c) SGG from re-implemented MOTIFS [71]. (d) SGG by the proposed unbiased prediction from the same model.

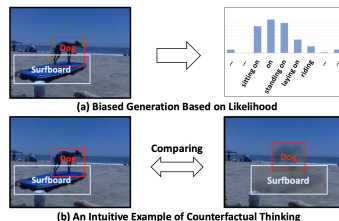
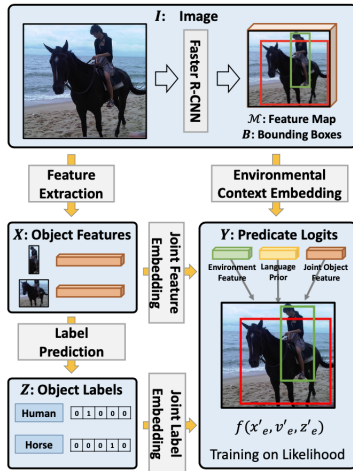
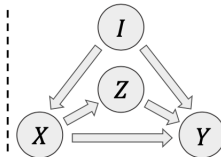


Figure 2. (a) The biased generation that directly predicts labels from likelihood. (b) An intuitive example of the proposed total direct effect, which calculates the difference between the real scene and the counterfactual one. Note that the “wipe-out” is only for the illustrative purpose but not considered as visual processing.

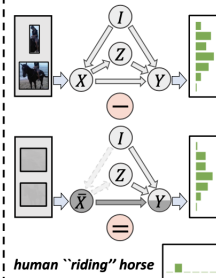
Application: Unbiased Scene Graph Generation (Tang et al. 2020)



(a) The SGG Framework Used for Biased Training



(b) The Causal Graph of SGG



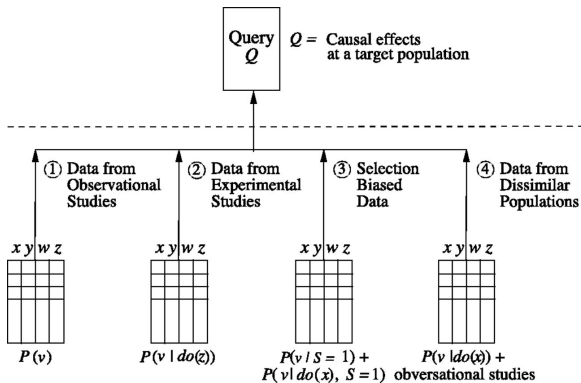
(c) Unbiased TDE Inference

Outline

- 1 Introduction
- 2 Why causal inference?
- 3 Causal graphs and Structural Equation Models
 - Causal graphs
 - Structural Equation Models (SEMs)
- 4 Related Topics and Applications
 - Building and Evaluation of Impact of a recommender system
 - Unbiased Scene Graph Generation
 - Data-fusion Problem

Application: Data-fusion Problem ⁹

- **Problem:** Piecing together multiple datasets collected under heterogeneous conditions (i.e., different populations, regimes, and sampling methods) to obtain valid answers to queries of interest.
- **Challenge:** the biases that emerge in heterogeneous environments.



⁹Elias Bareinboim and Judea Pearl. Causal inference and the data-fusion problem. PNAS, 2016.

References

-  Angrist, Joshua D., and Jörn-Steffen Pischke. Mostly harmless econometrics. Princeton university press, 2008.
-  M. Hernan and J. Robins. *Causal Inference*. Chapman & Hall/CRC, 2019.
-  S. Morgan and C. Winship. *Counterfactuals and Causal Inference*. Cambridge University Press, 2nd edition, 2015.
-  J. Pearl. *Causality*. Cambridge University Press, 2nd edition, 2009.
-  J. Pearl. Causal inference in statistics: An overview. Statistical Surveys, 2009.
-  C. Shalizi. *Advanced Data Analysis from an Elementary Point of View*. In preparation, 2019.